

Individual differences in the computational processes of face recognition and their association to face recognition ability

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1. Introduction

Paragraph 1: general comment on face recognition and on the important variability in the “normal” population

Being able to quickly and accurately identify a face is blabla in every day’s life. This ability becomes even more important in several real-life task, like eye-witness identification or passport control. However, face recognition ability is not distributed equally across the population. Without organic cause (like brain damage), people affected with developmental prosopagnosia show a poor face recognition ability throughout their life (Dobel et al., 2007; McConachie, 1976) and report an inability to recognize familiar faces. The other end of the face recognition ability spectrum regroups the “super-recognizers”. These individuals show a remarkably high face recognition ability compared to the average person (Bobak et al., 2016; Russell et al., 2009). However, even when ignoring these two extreme cases, face recognition ability remains highly variable in the normal population, especially for unfamiliar faces recognition (Dowsett & Burton, 2015; White et al., 2014). In this project, by linking experimental, differential and computational approaches, we tried to build on results from one of our previous work to investigate how individual differences in face recognition ability in the normal population is related to computational processes that support face recognition.

Paragraph 2: past investigation of individual difference in face identity processing

Differential psychology can be understood as the “study of the normative individual differences in thought, behavior and ability” (Marsh & Boag, 2013). Multiple fields of theoretical and applied psychology have used a differential approach to understand variability within the general population including psychometric assessment, educational and developmental psychology, and even psychopathology (Lubinski, 2000). Regarding face recognition, there is a growing interest in the value of investigating individual differences across the entire spectrum of face recognition ability as well, and not only by using group-level comparisons between the two extremes (prosopagnosic patients and “super-recognizers”) and average recognizers (White & Kitchen, 2022). Several tests have been designed to measure performance on defined set of tasks like new faces learning (Cambridge Face Memory test, CFMT; Duchaine & Nakayama (2006)), unfamiliar face perception (Glasgow Face Matching Test, GFMT; Burton et al. (2010)) or familiar face perception (Before They Were Famous, BTWF; Russell et al. (2009)). Using correlational approaches, past research has derived quantitative measures of the association between the performances at these tests or between test scores and other broader perceptual and cognitive processes (McCaffery et al., 2018; Noyes et al., 2018). This individual difference approach provides an opportunity to investigate the relationships between fundamental processes that are difficult to uncover using group-level analyses.

Paragraph 3: key question: domain-specific vs domain-general

For example, one key question that association studies have investigated is to what extend individual difference in face processing reflect domain-general differences in other perceptual and cognitive abilities (McCaffery et al., 2018; Verhallen et al., 2017). Using tests scores as variables of interest, some studies have shown that face and general object perception ability are positively associated but with modest effect sizes (Dennett et al., 2012a; Richler et al., n.d.; Wilmer et al., 2012). However, these associations remains constantly weaker than those between different face or object recognition task (Fysh et al., 2020; McCaffery et al., 2018; Wilmer et al., 2010), suggesting that domain-specific processing explains at least some part of the variance between these tasks (Richler et al., n.d.).

Paragraph 4: Investigating individual differences in latent processes using computational approaches

While the associations between manifest measures like test scores remains important to understand relationship between fundamental processes, performance remains a distant proxy of the underlying cognitive processes at work during face recognition, thus could not be ideal for capturing individual differences in these processes (Haines et al., n.d.). Using a computational approach to disentangle the multiple components of these processes could provide meaningful information regarding the relationships between the specific parts of the face recognition process

and face recognition ability (Yarkoni, 2022). Analyzing individual differences in these components could provide meaningful insights about cognitive variability in general as well and as how variation in these latent unobserved processes, which are driving behavior, is related to individual face recognition ability (White et al., 2016).

Paragraph 5: building upon our past project results summary and application of individual differences in face recognition ability

One of our previous work (Fournier, Downing & Ramsey, in press) has used Bayesian hierarchical evidence-accumulation models (see Figure 1) to fit choice-rt data in a face recognition (old/new) task (for a review on evidence-accumulation models see Ratcliff et al. (2016)). We found that distinct computational processes involved in the decision-making process of face recognition are affected by face familiarity. Compared to learned faces, novel faces induced a great accumulation of evidence (drift rate) and the response caution (threshold) towards the novel response was higher as well. However these conclusions were drawn from an experimental psychology approach (i.e. from the group-average posterior distributions of these two parameters) and this approach did not consider the fact that the between-subject variability in these estimated parameters was important. In the present study, we used an individual difference approach to investigate the between-subject variability found in these estimated parameters. More specifically, we were interested to see if the estimated parameters computed from choice-rt data in a face memory task were associated with face recognition ability specifically or if they reflect a general recognition ability unspecific to faces. We think that this approach could give precious insights about what do the parameters reflect and how specific to face recognition the inferences which are drawn from their investigation are.

Paragraph 6: Short description of how we are going to address the questions

We used a similar face memory task to the one described in Fournier et al. (in press) to compute the threshold and drift rate parameter for both learned and novel faces. In addition to this task, face recognition ability was assessed using the Cambridge Face Memory Test (CFMT) and general object recognition ability was assessed by using the Cambridge Car Memory Test (CCMT). We wanted to investigate to what extent do drift rate and threshold vary as a function of face recognition ability alone and when adjusted by general object recognition ability. Based on our pilot work, we hypothesized that the accumulation of evidence for learned faces when (drift rate learned-TRUE)

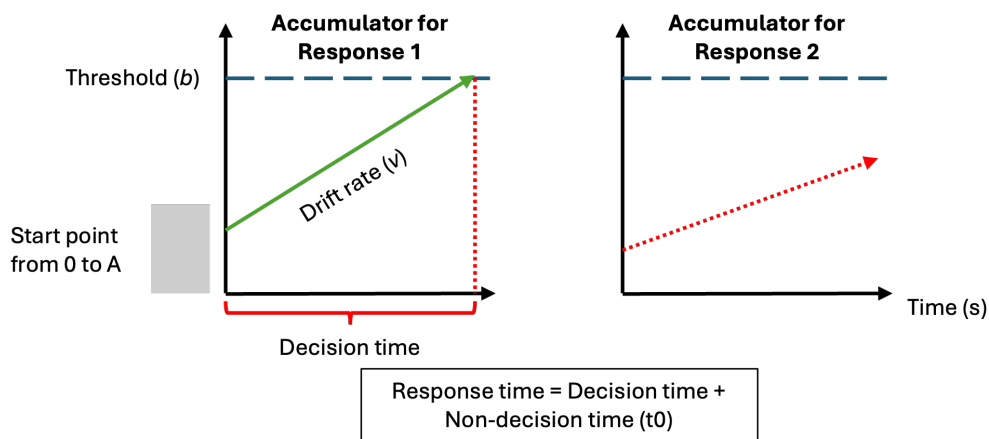


Figure 1. Schematic of the linear ballistic accumulator (LBA) model

This diagram represents the accumulation process one trial of a perceptual decision task for the linear ballistic accumulator (LBA; Brown & Heathcote (2008)). Participants are required to choose between two options (Response 1 and Response 2). In a LBA model, there is a separated accumulation of evidence (v) for each response option. In this example, the correct response is Response 1. The accumulator on the left is the accumulator for the correct response (Response 1), and the green line in represents the rate at which the evidence accumulates in favor of that response option. The accumulator on the right is the accumulator for the incorrect response (Response 2), and the dash red line represents the rate at which the evidence accumulates in favor of this response option. Once the accumulated evidence crosses the threshold (b) of a response option, the corresponding response is chosen (in this case Response 1).

will be positively associated with face recognition ability and that this association will remain after adjusting for general object recognition.

2. Method and Materials

2.1. Pre-registration and open science

We report how we determined our sample size, all data exclusions (if any), all manipulations, and all measures in the study (Simons et al., 2017). The research question, hypotheses, experimental design, planned analysis, and exclusion criteria were preregistered (<https://osf.io/b74gt/overview>). The raw data, experimental materials, and analysis code are available on GitHub (add link) or the Open Science Framework (add link).

2.2. Participants

All participant were recruited on the Prolific platform (<https://www.prolific.com>). Participants had to be located in the United Kingdom or in the United States, be aged between 18-65 years old, with normal or corrected-to-normal vision and speak fluent English. Furthermore, Prolific offers the option to chose the approval rate of the participants willing to take part in the study. The approval rate is the percentage for which participants has been approved in previous experiment. An approval rating between 95% and 100% was chosen in our study to try to ensure high data quality. 383 participants completed the experiment. Following our preregistered exclusion criterion, 2 participants were excluded because their percentage of missed trials was higher than 40%, 3 because their percentage of RT outside the established range (0.2 to 2 seconds) were higher than 25% and 78 were removed because the overall accuracy was lower than 55%. In the end, data from 300 participants (141 men and 159 women) were retained for the analysis and the LBA modeling. Ages ranged from 19 to 64 ($M_{age} = 42$, $SD_{age} = 11.2$). Following Prolific payment principles, each participant received 9£ for the completion of the study. This experiment was accepted by the ethics committee of the Federal Institute of technology (ETH) Zurich.

2.3. Material

Participant were asked to complete the study using a desktop computer. Beforehand, the paradigm was tested on three different operative systems (PC, Macintosh and Linux) and on three web browsers (Google Chrome, Safari and Firefox). Participant were aware that using any other combination of operative systems and web browsers might lead to the inability to complete the experiment and were asked to follow these directives to ensure a consistent procedure and participant experience.

2.4. Procedure

The experiment was divided into three experimental tasks. A face recognition task was used to estimate the LBA parameters from manifest measures (accuracy and response times). Face and car recognition ability were assessed using the Cambridge Face Memory Test and the Cambridge Car Memory Test. Participants always first underwent the face recognition task and the order of two memory tests was counterbalanced.

2.4.1. Face recognition task

The face recognition task was divided into two distinct parts: a familiarization part and a recognition part. During the familiarization part, unknown identities were presented on the screen and participants were instructed to memorize the identity of the faces. One familiarization trial consisted of the presentation of 6 face images, one after the other, showing the same identity but varying in viewing points (3/4 left view, frontal view and 3/4 right view) and the colorimetry (colorful image, gray scale image). The duration of each image presentation was set to 0.75 second and each participant underwent 30 familiarization trials (for 30 unknown identities). During the familiarization, we introduced catch trials to ensure participants attention. After some familiarization trials, a gray scale image of a person's face was presented on the screen. Participants were asked to press the "q" key if the face belonged to the last seen identity or the "p" key if it was

not. The faces seen during the familiarization were considered the “Learned faces” within-subject condition.

In the recognition part, participants completed three blocks of 90 recognition trials. Each block was separated from another by a short break. Half of the trials were learned identities trials and the other half were new, previously unseen identities trials. In each trial, a gray scale image of a face was shown on the screen. Participants were asked to choose if they had previously seen this face in the familiarization part or not. If they did, they should press the “q” key and if they did not, the “p” key. Participants were asked to respond as quickly and accurately as possible. It is important to note that participants never saw two pictures of the same learned identity in the same block and never saw the same picture (3/4 left view, frontal view, or 3/4 right view) of a learned identity during the entire recognition part. Furthermore, each novel identity was only presented once in the entire recognition part. Response time [s] and accuracy were collected after each recognition trial. A response time limit of 2 seconds has been set for each recognition trial. If participants took longer than 2 seconds to press a response key, the text “Too late!” appeared in red on the screen and the trial was discarded, but not replaced.

2.4.2. Cambridge Memory Tests

The method of the CFMT was described in [Duchaine & Nakayama \(2006\)](#). Participants learned 6 Caucasian male faces in three views and had to discriminate them later from distractor faces (no time response limit, three-alternative forced-choice task, Figure 3 a). Regarding the CCMT, the method has been described in [Dennett et al. \(2012a\)](#) and has a highly similar structure, format and scoring as the CFMT, but uses cars instead of faces as stimuli.

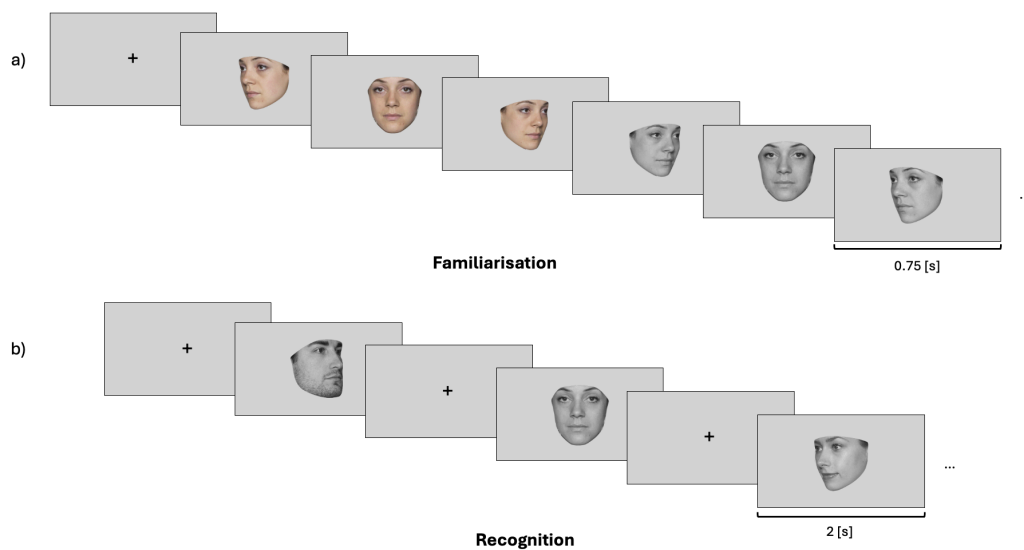


Figure 2. Experimental procedure of the face recognition task

The experimental procedure was divided into two parts: a familiarization part and a recognition part. a) In the familiarization part, participants were asked to memorize the face identity of 30 identities. For each identity, participants looked at 6 pictures depicting the same identity with different face orientation (3/4 left view, frontal view and 3/4 right view) and different colorimetry (color and gray scale images) in the order presented here: 3/4 left view, frontal view and 3/4 right view in color, followed by 3/4 right view, frontal view and 3/4 left view in gray scale. b) In each trial of the recognition part, participants had to decide if they recognize the face shown on the screen from the familiarization part. If they did, they had to press the “q” key and if they did not, the “p” key. They completed three blocks of 60 trials (half of them were from the familiarization part, half were novel faces). For the recognition part, only gray scale images of faces were used (for both learned faces and novel faces).

2.5. Stimuli

The face images used in the faces recognition task consisted of images selected from two different data sets of face images: the Face Research Lab London Set (DeBruine, Lisa; Jones, Benedict (2017): <https://doi.org/10.6084/m9.figshare.5047666.v5>) and the Color FERET Database (<https://www.nist.gov/itl/products-and-services/color-feret-database>). Each of these data sets is composed of images of different identities faces taken from different angle points. 102 identities were selected from the Face Research Lab London Set and 51 from the Color FERET Database. For each selected identity, three pictures of the face at three different view points (3/4 left view, frontal view and 3/4 right view) were cropped following the outline of the face.

To ensure that the stimuli that we used are reproducible, we used the webmorphR R Package (version?) -> NEED info from Rich here.

2.6. Evidence accumulation modeling

The model and priors specification as well as the model estimation are all identical to the one described in (ref: Fournier et al., in press).

2.6.1. Model specification

A hierarchical LBA models was fit to the accuracy and response times of the face recognition task. The model specifications are outlined in our preregistration (Table 1). The LBA framework assumes a separate evidence accumulator for each response alternative, with potentially different parameter values. Accordingly, one accumulator corresponded to the “Learned” response and the other to the “Novel” response. Threshold was allowed to vary by participants’ response (Learned/Novel)

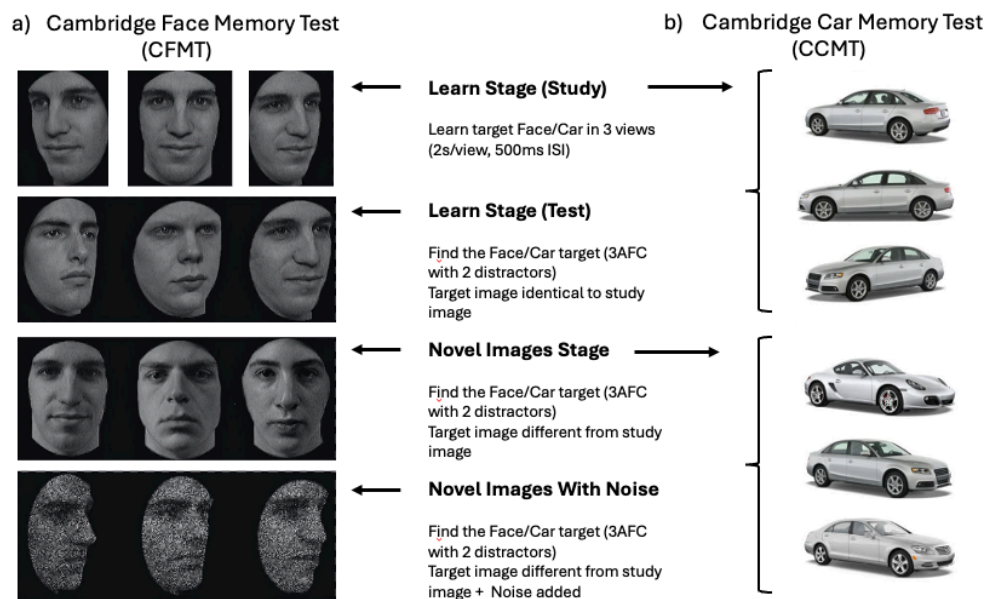


Figure 3. The two Cambridge Memory Tests

Illustration of (a) the Cambridge Face Memory Test (CFMT; ref Duchain & Nakayama, 2006) and (b) the Cambridge Car Memory Test (CCMT; red Dennett et al., 2012a). These two tests are highly similar, differing only in the category of the presented stimuli. Both tests are composed of three stages: Learn (including Study and Test), Novel Images and Novel Images with Noise. In the Learn stage, participants learn target faces/cars and are then tested on the recognition of the identical images to the learned targets. In the Novel Images and Novel Images with Noise stages, recognition targets are different pictures of the same learned targets, but differing in views, lightning or both. For the CCMT (b), the car pictures showed here are not the one used in the CCMT and served only as a representation of the stimuli used in the Learn stage and the Novel stages of the CCMT. This illustration has been taken and slightly modified from Dennett et al. (2012b). 3AFC = three-alternative forced choice; ISI = interstimulus interval.

and mean drift rate varied as a function of both stimulus type (Learned/Novel) and whether the accumulator corresponded to the correct response for that stimulus (TRUE vs. FALSE). For example, when a Learned stimulus was presented, the Learned accumulator was designed as TRUE, whereas the Novel accumulator was designed as FALSE. The difference in drift rate between the TRUE and FALSE accumulators reflects the quality of information supporting a decision, with larger differences indicating stronger and more discriminative evidence accumulation (Lewandowsky & Oberauer (2018)). The standard deviation of the TRUE accumulator was allowed to vary according to accumulator match status, whereas the standard deviation of the FALSE accumulator was constrained to be constant to ensure model identifiability (Donkin et al. (2009)). In the preregistered model, a single start-point parameter (A), and single non-decision time parameter (t0) were estimated. Alternative model specifications that permitted variation in these parameters are reported in the Supplementary Materials.

Table 1. Formula specification for the pre-registered linear ballistic accumulator model.

Parameter	Label	Condition	Response	Accuracy	Fixed
Start point	A	-	-	-	1
Mean drift rate	v	yes	no	yes	-
SD drift rate	sd_v	no	no	yes	-
SD false drift rate	false.sd_v	-	-	-	1
Threshold	b	yes	yes	no	-
Non-decision time	t0	yes	no	no	-
SD Non-decision time	st_0	-	-	-	0

Condition = training condition (Learned vs. Novel); Response = key-press response (same vs. different); Accuracy = (true/correct vs false/incorrect); SD = standard deviation.

2.6.2. Model estimation

Bayesian model estimated was carried using the EMC2 R package (Stevenson et al., 2026). Priors were chosen to be weakly-informative, meaning that they have little influence on the estimation (for more details regarding the priors of each model, see Supplementary Materials). Two sampling steps were used to estimate the parameters. First, a separate estimation of the fixed effects for each participant was conducted. Parameter estimations were then used as the starting point for the full hierarchical parameter estimation.

2.6.3. Model fit

To ensure that the fit of our model can capture most of the main trends in the data, graphical summaries of the posterior predictions were examined. (Check fit of the model, Supplementary Materials)

2.7. Data analysis

Regarding the general analytical strategy, two main approaches were considered. Correlations and data visualizations were used to characterize the zero-order relationships between parameters of interest (drift rate and threshold) from the LBA model with face and object recognition ability. Linear regression was then conducted to estimate how much LBA parameters vary as a function of face recognition ability, as well as how much these parameters vary after adjusting for car recognition ability. The details of the analytical strategy were preregistered (<https://osf.io/b74gt/overview>).

2.7.1. LBA parameters

In order to make inference about the magnitude of familiarity effects, differences in parameter estimates across conditions were compared. To summarize the distribution of parameters of interest, we report the median of the posterior distribution together with 95% quantile interval of the posterior distribution.

2.7.2. Classical correlational analysis

The median value of each parameter's posterior distribution was correlated with the two centred scores of each of the Cambridge Memory Tests to assess the relationship between these variables.

2.7.3. Bayesian plausible value correlational analysis

In addition to the classical correlational analysis described above, we conducted a Bayesian plausible value correlation approach (Ly et al., 2017; Matzke et al., 2017). Instead of using point estimates (median value of each parameter's posterior distribution), this approach permits the estimation of uncertainty in likely correlational values by correlating the parameter estimation from each sample (or "draw") from the posterior distribution with the memory scores. A distribution of the plausible correlations between each parameter and the memory scores can then be obtained. These distributions can be compared between stimuli conditions (Learned vs Novel) or memory scores (face vs car) using difference contrasts. It allowed us to estimate if the strength of the correlation differs or not between conditions of interest.

2.7.4. Linear regression

Based on our pilot results, we preregistered two Bayesian linear regression models to quantify the extent to which drift rate varies as a function of face recognition ability alone and when accounting for car recognition ability. The results of the linear regression for the threshold parameter are available in the Supplementary materials. An index coding approach was used (no grand intercept) where each condition of interest has its own intercept (see Table 2 for the details of each regression models).

Table 2. Formula specification for the pre-registered linear regression models.

Parameter	Condition	Ability	Formula
Median drift rate	familiarity:accuracy	Face	Median drift rate = 0 + familiarity:accuracy + face_ability + (0 participant)
Median drift rate	familiarity:accuracy	Face + Car	Median drift rate = 0 + familiarity:accuracy + face_ability + car_ability + (0 participant)

familiarity = face familiarity (Learned vs. Novel); accuracy = response accuracy (TRUE vs. FALSE)

2.7.5. Inferential criteria

Using a Bayesian approach in our regression analysis allows us to evaluate the posterior distributions of the parameter of our predictor of interest (Face Memory score). To evaluate these posterior distributions, we will use three complementary criteria: sign, size and precision (Cumming, 2014; Kruschke & Liddell, 2018; McElreath, n.d.). To address the sign (or direction) of the effect, we will report if the median value is positive or negative. Second, we will evaluate the size or magnitude of the posterior distribution by considering the size of the median value as well as the extent to which lower or upper bounds of the 95% quantile interval approach zero (no relationship). For example, if the 95% quantile interval excludes zero, we could be reasonably confident in our inference of a non-zero effect. Furthermore, we will use the width of the 95% quantile intervals as an indicator of estimation precision. Whereas narrow estimates indicate precise estimates, wider intervals indicate substantial uncertainty about the estimated parameter value, which may warrant a more cautious interpretation.

3. Results

3.1. LBA parameters

Posterior distributions for drift rate and threshold parameters are visualized in Figure 4 and Figure 5 and the medians and 95% quantile intervals are summarised in Table 3.

Threshold: Inspection of the posterior distribution shows that there is an effect of familiarity on the threshold parameter.

Drift rate: Inspection of the posterior distribution shows that there is a familiarity effect in drift rate difference. The quality of the evidence accumulation was greater for Novel faces compared to Learned faces.

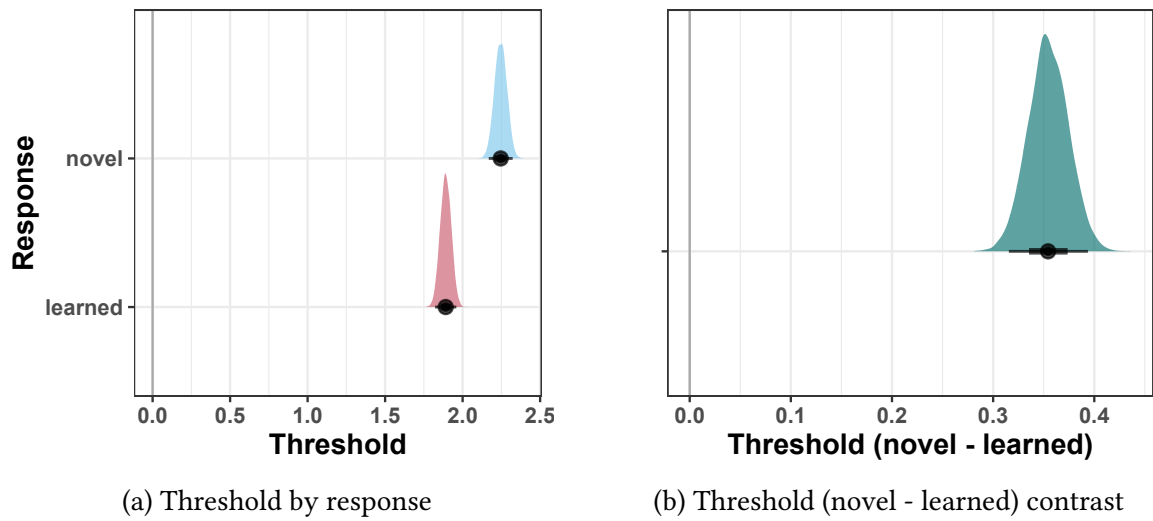


Figure 4. Threshold estimates

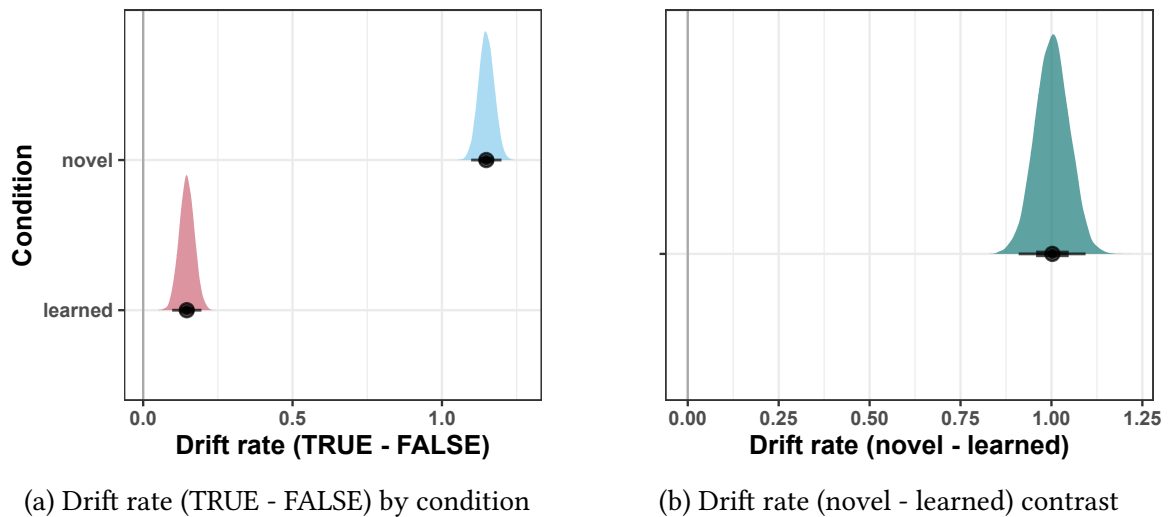


Figure 5. Drift rate estimates

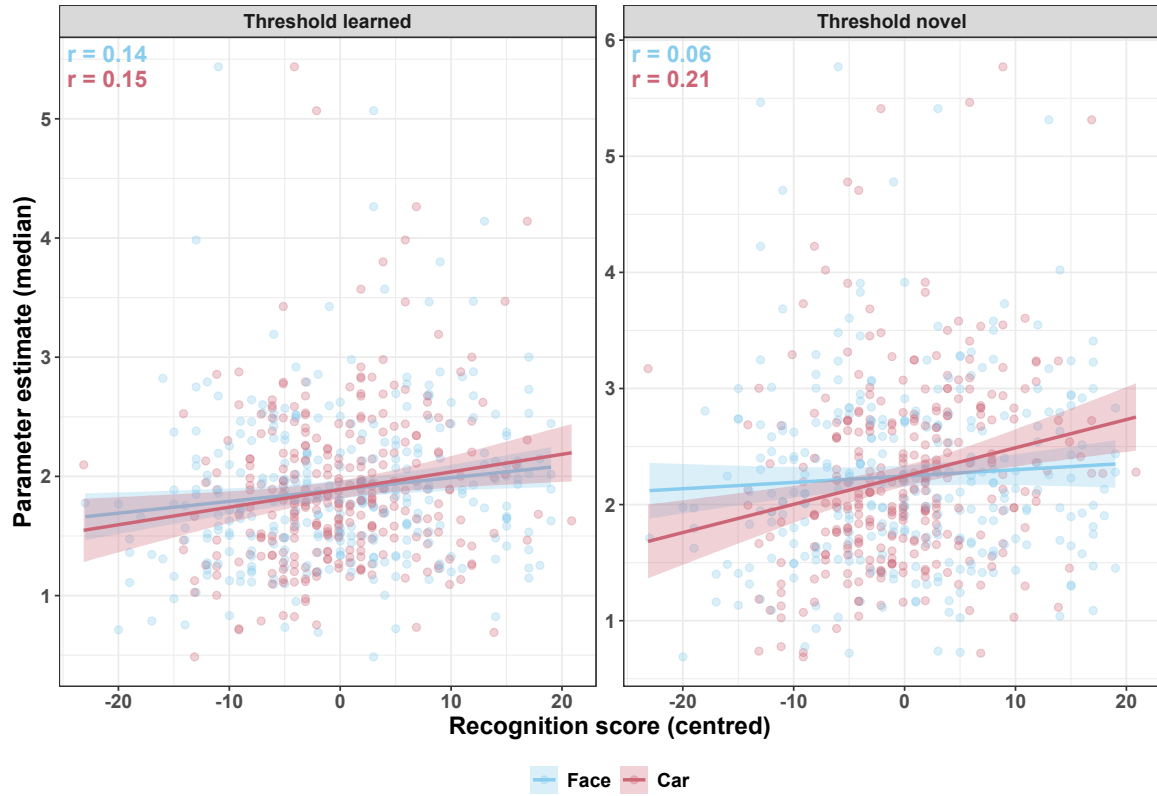
Table 3. Posterior median and 95% quantile intervals for threshold and drift rate parameters.

Parameter	Condition	Median	95% QI
Threshold	novel	2.25	[2.169, 2.323]
Threshold	learned	1.89	[1.822, 1.96]
Threshold contrast	novel - learned	0.35	[0.316, 0.394]
Drift rate	novel	1.15	[1.098, 1.199]
Drift rate	learned	0.15	[0.097, 0.195]
Drift rate contrast	novel - learned	1	[0.91, 1.094]

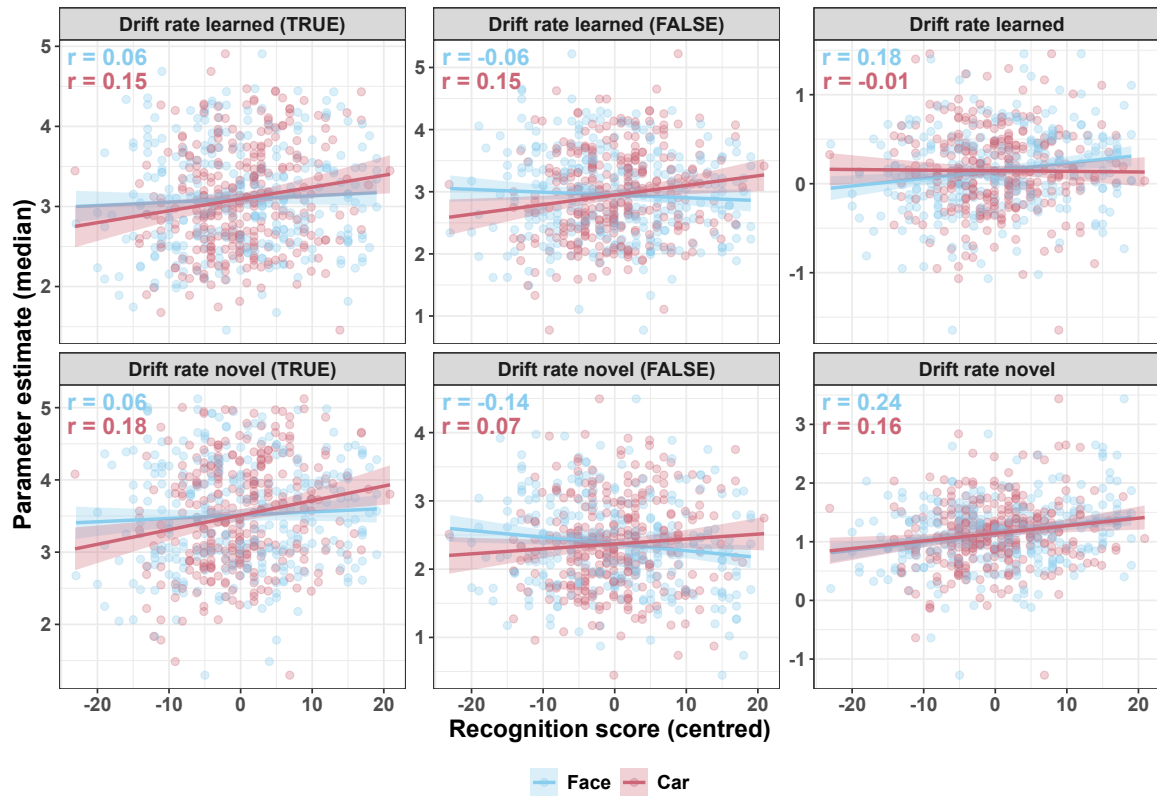
QI = quantile interval.

3.2. Correlations

We examined point-wise recognition ability linear relationship with point estimates (median) of the posterior distribution of each of the LBA parameters of interest. These relationships can be seen in Figure 6 for both recognition ability. Visual inspections of these correlations seemed to show that the accumulation of evidence (drift rate) for learned and novel faces increases as face recognition ability increases. A similar conclusion could be drawn from the inspection of the response caution (threshold) towards the “Learned” response. However the strength of these relationships remains weak (the highest r value is 0.24) and we used a Bayesian plausible value correlational approach to determine the uncertainty linked to these associations.



(a) Median threshold correlations with recognition ability



(b) Median drift rate correlations with recognition ability

Figure 6. Point estimates correlations with recognition scores

3.3. Bayesian plausible value correlational analysis

To estimate the uncertainty linked to the relationships between LBA estimated parameters and face recognition ability, we used a Bayesian plausible value correlational analysis. Like described above (see Methods), instead of using a single point estimate to assess the strength of the relationship between our variables of interest, the posterior distribution of the LBA parameters allow us to use the entire sampling distribution to do it. We calculated the Person's r between each sample value from our posterior distribution for each parameter of each subject and their face recognition ability. The posterior distributions of threshold plausible correlation values and their contrast are shown in Figure 7. Posterior correlations with true and false drift rate for both conditions and recognition abilities can be found in Figure 8, and correlations with drift rate differences (true-false) by conditions as well as drift rate contrast (novel-false) in Figure 9. Table 4 provides a summary of these distributions.

Threshold: There is a positive relationship between response caution for both learned and novel faces and recognition ability. Regarding face recognition ability, people with a higher face recognition ability have a higher response caution for learned faces whereas people with a higher car recognition ability have a higher threshold for both learned and novel faces.

Drift rate: A higher face recognition ability is associated with a greater drift rate for both novel and learned faces. However, a higher car recognition ability is only associated with an increase drift rate for novel faces.

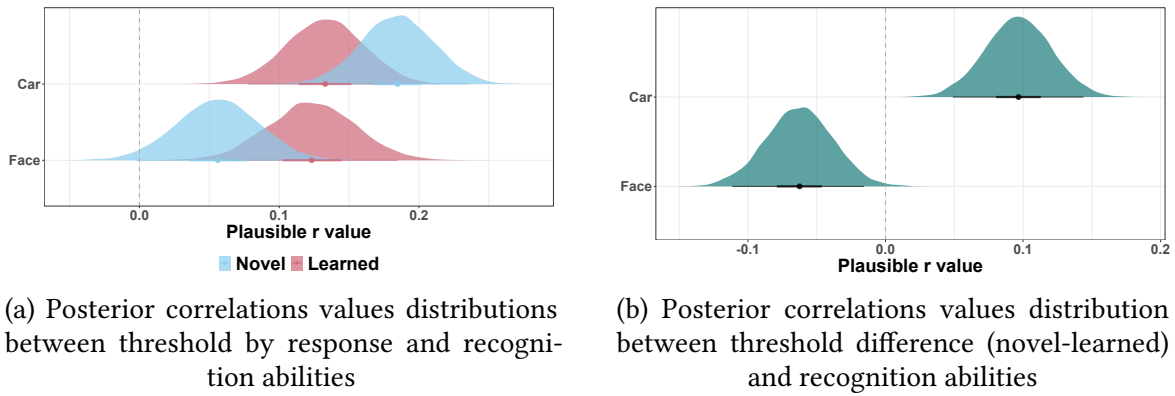


Figure 7. Plausible correlation values distributions between threshold and recognition abilities. Points denote the median of the distribution and error bars represent 95% quantile intervals.

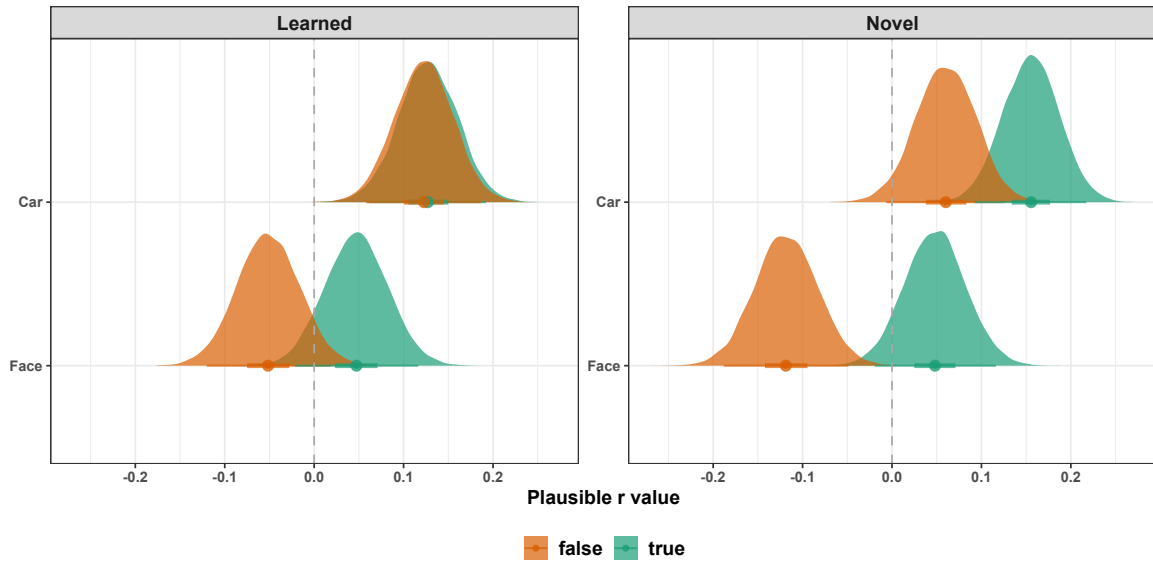
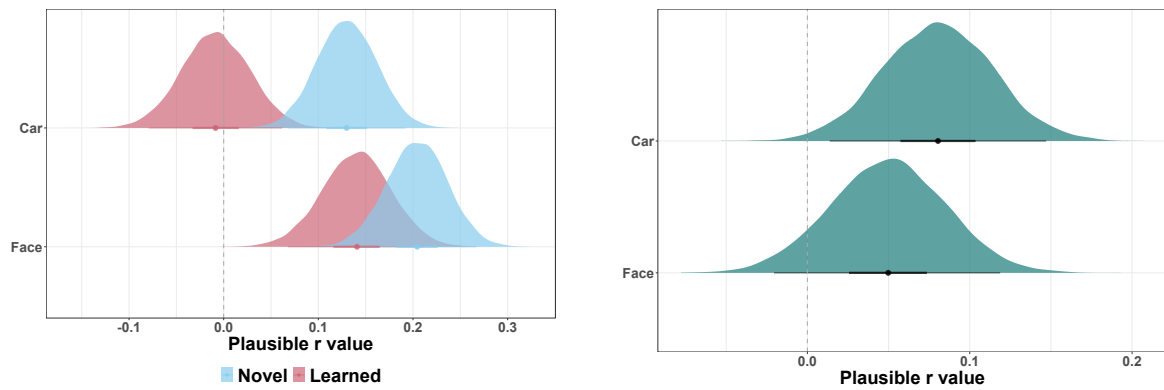


Figure 8. Plausible correlation values distributions between drift rate TRUE and FALSE by stimulus and recognition abilities. Points denote the median of the distribution and error bars represent 95% quantile intervals.



(a) Posterior correlations values distributions between drift rate by stimulus (TRUE - FALSE) and recognition abilities

(b) Posterior correlations values distribution between drift rate difference (novel-learned) and recognition abilities

Figure 9. Plausible correlation values distributions between drift rate differences and recognition abilities. Points denote the median of the distribution and error bars represent 95% quantile intervals.

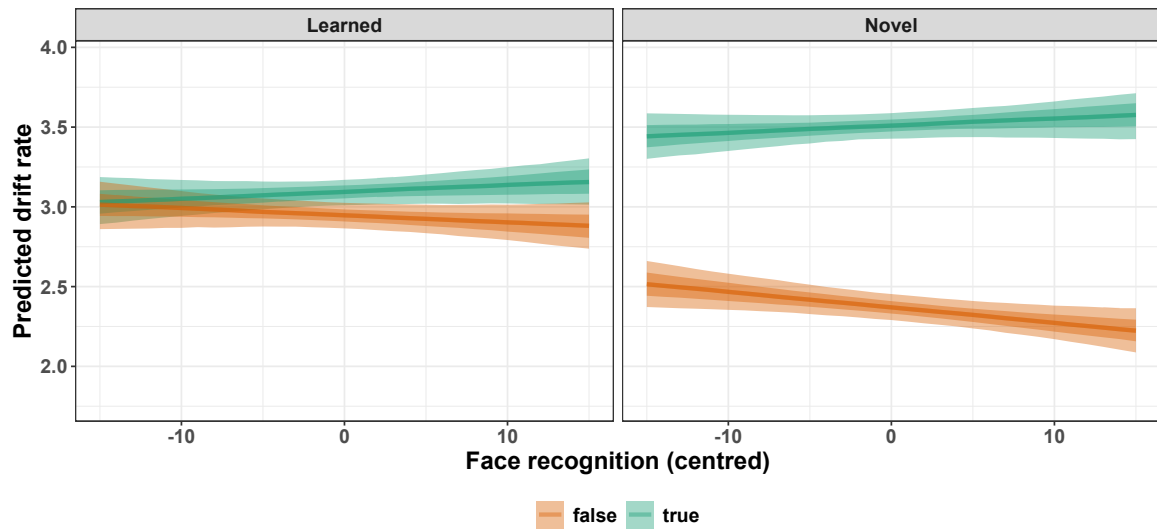
Table 4. Median r and 95% quantile intervals of the plausible value correlations between LBA parameters and recognition ability scores.

Parameter	Condition	Accuracy	Face		Car	
			median r_{Face}	95% QI_Face	median r_{Car}	95% QI_Car
Threshold	Learned	-	0.12	[0.06, 0.18]	0.13	[0.08, 0.19]
Threshold	Novel	-	0.06	[-0.00, 0.11]	0.18	[0.13, 0.24]
Threshold	Novel-Learned	-	-0.06	[-0.11, -0.02]	0.10	[0.05, 0.14]
Drift	Learned	True	0.05	[-0.02, 0.12]	0.13	[0.06, 0.19]
Drift	Learned	False	-0.05	[-0.12, 0.02]	0.12	[0.06, 0.19]
Drift	Learned	-	0.14	[0.07, 0.21]	-0.01	[-0.08, 0.06]
Drift	Novel	True	0.05	[-0.02, 0.12]	0.16	[0.09, 0.22]
Drift	Novel	False	-0.12	[-0.19, -0.05]	0.06	[-0.01, 0.13]
Drift	Novel	-	0.20	[0.14, 0.27]	0.13	[0.07, 0.19]
Drift	Novel-Learned	-	0.05	[-0.02, 0.12]	0.08	[0.01, 0.15]

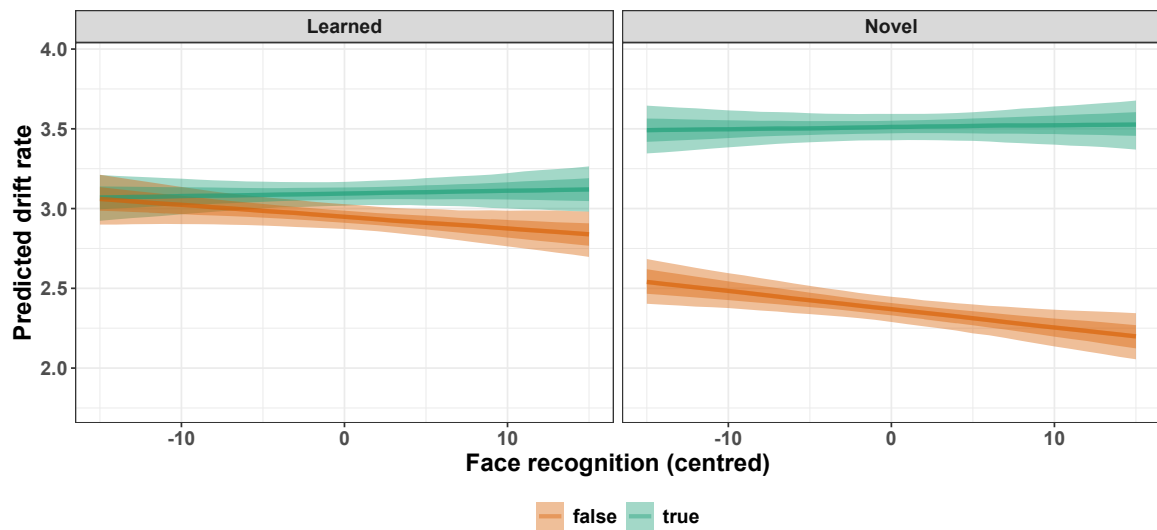
Note. Blue = 95% QI entirely above zero; red = 95% QI entirely below zero.

3.4. Bayesian linear regression

Following our preregistration, the predictions of the two regression models for drift rate can be seen in Figure 10.



(a) Face model: predicted drift rate (TRUE and FALSE) values from face recognition ability scores



(b) Face and Car model: predicted drift rate (TRUE and FALSE) values from face recognition ability scores when accounting for car recognition ability

Figure 10. Predicted drift rate values (TRUE and FALSE) from face recognition ability scores

4. Discussion

Paragraph 1: Opening paragraph

The aim of this project was to combine experimental, differential and computational approaches to investigate the relationship between face recognition ability and the computational mechanisms involved in the face recognition process. Our results showed that individual differences in face recognition ability are not only related with different latent computational processes (and not only one), but also predict to a certain extend the pattern of these processes. Face recognition ability seems to be positively associated with the rate of evidence accumulation (drift rate parameter) as well as the response caution (threshold parameter) for both learned and novel faces recognition. One key finding is that these predictions still hold when accounting for object

recognition ability, demonstrating that these these associations are partially specific to faces and cannot entirely be explained by a more general object recognition ability.

Paragraph 2: Drift rate and threshold effects

One important consideration is the fact that the effect of individual differences in face recognition ability on the LBA parameters is multifaceted. Drift rate and threshold are both positively associated with face recognition ability. Our drift rate results suggest that people who are generally better at recognizing faces accumulates more recognition signal and are more capable to ignore noisy information coming from the stimulus. Our findings are in accordance with previous research that showed that expertise effects are primarily due to a better information extraction (Litchfield & Donovan, 2017; Thompson et al., 2013; Van Der Gijp et al., 2014). We extend these findings by showing that these effects are not limited to group-level differences between experts and novices but concern the range of the “normal” population as well. Regarding response caution, previous studies have shown that confidence is itself a decision variable used to regulate future behavior (Fleming et al., 2018; Folke et al., 2016). The positive association between threshold and face recognition ability could reflect a better estimation of the task difficulty for people with higher face recognition ability, who voluntarily increase their response caution to ensure a better performance given the task demands. However, future research is required to confirm this suggestion. Overall, our results suggest that not only does high face recognition ability predict a better ability to extract meaningful information for the recognition success but also predict the voluntary strategic decision of the subject to be more cautious.

Paragraph 3: Theoretical implications for individual differences investigation

Besides the insights obtained from the associations between individual differences and latent computational processes in the domain of face recognition, our choice of combining experimental, differential and computational methodologies has several implications for future differential studies, as well as for models of face perception and memory. On a more general level, it offers a novel approach to investigate individual differences and expertise effects in face processing. Most differential studies in face processing have used test scores to estimate different abilities underlying face processing and investigated if these abilities were related to each other or to other predictors of interest (McCaffery et al., 2018; Noyes et al., 2018). In this study we used traditional approach of an experimental manipulation in a large sample, extended the separate analysis of response times and accuracy to the estimation of latent variables through computational modelling that we then associated with face recognition ability. The important difference with past investigations is that behavioral patterns (accuracy and response times) as well as latent variables patterns (LBA parameters) were associated with ability scores. Using this approach allowed more specific hypotheses regarding which parts of the system is related to which type of recognition abilities, potentially aiding the development of theory. We think that this approach could benefit future research on individual differences in visual recognition memory or visual perception more generally.

Paragraph 4: Theoretical implications for models of face perception and memory

The insights obtained in this study also have theoretical implications when interpreting them in accordance with existing models of face processing system. Existing models of the human face processing system propose different cognitive and neural pathways (Bruce & Young, 1986) for the general processing face... Previous studies findings have increased our understanding of the general process by showing...

- Bruce and Young model
 - Individual differences in face recognition: observations from comparing group-levels difference between two different populations (ex: prosopagnosia and control/super-recognizer and control) / individual differences in typical range of face processing ability
- While Bruce and Young model describe the face perception process in general, there are some individual variabilities in the outcome of this process (outcome = performance) and in other steps in this □ With this study we also contribute to the extension of this model by showing that not only there are individual differences in the outcome, but that there is also individual differences in the cognitive processes that give rise to this outcome. □ Future research could use an individual differences approach to investigate

how some variations in the components in the Bruce and Young face recognition model could influence estimated latent variables and how are individuals differences in face recognition ability are related to the observed differences of latent variables of interest.

Paragraph 5: Mentioning the object recognition ability correlations

While no hypotheses regarding the associations between the LBA parameters and the object recognition ability was preregistered, the exploratory analysis of these associations reveal interesting patterns. Object recognition ability is positively correlated with drift face for novel faces but not for learned faces. Furthermore, object recognition does not predict the discrimination between TRUE and FALSE drift rate for learned faces like face recognition ability does. Object recognition is known to rely mostly on individual features processing while face recognition combines featural and configural processing (Cabeza & Kato, 2000; Madole & Cohen, 1995; Tversky & Hemenway, 1984). Novel faces recognition relies more on feature information matching and, as a novel face becomes more familiar, the reliance on configural processing increases (Schwaninger et al., 2002). We suggest that relying on visual feature processing alone help to accumulate meaningful information for the recognition of visually familiar faces but also contribute to the accumulation of information detrimental to the recognition process (noise). Whereas relying on both (visual information processing and holistic processing) seems to increase the accumulation of meaningful information while decreasing the noise coming from detrimental visual information to the recognition process. Future research should investigate if the different patterns of associations between recognition abilities and latent components of the recognition process are related to different types of information processing.

Paragraph 6: Limitations

Effect size: Overall, the effect size of the different associations interpreted above are small, even if the posterior plausible values correlation distributions don't overlap with 0...

Regarding the LBA parameter estimation, previous evidence accumulation model development studies have highlighted how the number of trials per cell of the design influence parameter estimation (Alexandrowicz & Gula, 2020; Lerche et al., 2017; Wagenmakers et al., 2007). Our design (90 trials per condition) and control of error rates in our face recognition task followed the minimum recommendations to apply EAM (Lüken et al., 2025). Nonetheless, adding more trials per condition (e.g., 200 per condition; Boag et al. (2025)) would have increased the robustness of our parameter estimates.

5. Conclusion

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